Week 14

Course Level Objectives:

- Perform hypothesis testing using statistical methods. (CLO 8)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. State the hypotheses for a chi-square test of association from a two-way table. (CLO 8) **2.** Compute the expected counts for a two-way table. (CLO 8) **3.** Check that all expected counts are at least 5. (CLO 8) **4.** Compute the chi-square statistic and its degrees of freedom, $(r - 1)(c - 1)$. (CLO 8) **5.** Interpret the right-tail p-value of a chi-square test. (CLO 8) **6.** Relate the chi-square test on a 2×2 table to the two-proportion z test. (CLO 8) **7.** Explain why a significant result shows association but not direction or causation. (CLO 8)	e-Text: 7.2 Testing for an Association Between Two Categorical Variables Interactive Lecture: 7.2 Testing for an Association Between Two Categorical Variables Slides: 7.2 Testing for an Association Between Two Categorical Variables StatKey – Theoretical Distributions (χ^2)	Participation: Chi-Square Test Your Turn: Faculty Rank and Gender: compute expected counts, cell contributions, χ^2 and df; decide and interpret	Online Homework: 7.2 Testing for an Association Between Two Categorical Variables (due Sun Nov 29, 11:59 PM)	WileyPLUS StatKey Blackboard

Also this week: Test 3 Retake – Mon Nov 23 – Sun Nov 29 (optional; higher score counts)